INSTRUCTOR GUIDE

How Do You Test Weather You Cannot Schedule?

A ready-to-teach guide for verifier-guided Best-of-N scaling. Includes 5, 15, and 30-to-40-minute routes.

Alicia Chua, Pawarit Laosunthara, and Eric Tang | Anyscale

The core lesson: More inference compute buys options. It does not buy judgment. A larger pool helps only when useful candidates exist and the selector can rank them reliably.

What this resource teaches

The lesson uses one controlled driving clip to make a broad generator-selector pattern visible:

1. Generate several candidates with a fixed model and condition.
2. Evaluate each candidate against an application-specific rubric.
3. Keep the highest-ranked eligible candidate, or abstain if none is acceptable.

Cosmos Transfer 2.5 is the case-study generator. Ray is the offline parallel execution layer. Neither is the concept.

Audience and prerequisites

- Level: first-year university through graduate study, plus technical practitioners.
- Needed: basic random sampling, seeds, weighted averages, and model conditioning.
- Not needed: diffusion mathematics, AV expertise, Python, Ray, CUDA, or a GPU.

Learning outcomes

- Explain Best-of-N as generation, scoring, and selection.
- Separate fixed conditions from seed-driven variation.
- Define observable criteria and apply hard rejection gates.
- Analyze compute, diversity, selector workload, and diminishing returns.
- Diagnose proxy optimization and rubric misalignment.
- Design a selector for another generative task.

Five-minute live route

0:00-0:45	**Hook**	A fleet can collect millions of ordinary miles and still miss the exact conditions it needs.
0:45-1:30	**Restage**	What must change, and what must remain recognizable?
1:30-2:15	**Blind vote**	Do not choose the prettiest clip. Choose the one you would trust as a test case.
2:15-3:15	**Scale N**	The task is fixed. Only the random seed changes.
3:15-4:20	**Reveal**	Did the generator fail, or did the selector ask the wrong question?
4:20-5:00	**Close**	Generation creates options. Selection makes the decision.

Preparation

- Open interactive/index.html and preload every clip.
- Press Reset. Confirm Take 04 is not identified before the learner submits a ranking.
- State that the generation has already been completed. Learners need no GPU.
- Give both a QR code and a typed URL. Keep the offline package ready.
- For the longer route, distribute the learner worksheet before revealing scores.

Accuracy note

These clips use edge conditioning. Depth is shown only as an alternative structural control. The displayed ratings are fixed illustrative annotations. They are not benchmark measurements or autonomous-vehicle safety claims.

Fifteen-minute guided route

0-2	**Missing cases**	Predict which conditions a fleet cannot collect on demand.	Long-tail coverage
2-4	**Transformation**	Compare source, edge, fog, rain, and snow.	Conditional restaging
4-6	**Blind vote**	Choose a clip and record visible evidence.	Implicit criteria
6-9	**Audition**	Expand N from 1 to 8 and rank the pool.	Inference-time search
9-12	**Selector lab**	Change criterion weights and watch rankings move.	Sensitivity and proxies
12-14	**Judge sources**	Assign roles to humans, metrics, VLMs, and gates.	Hybrid verification
14-15	**Check**	Answer two retrieval questions.	Mechanism and limits

Facilitation prompts

- Which visible property made you trust one candidate more?
- Were you choosing visual appeal or fitness for purpose?
- Whose priorities are encoded in these weights?
- What failure must never be averaged away?
- If a five-point weight change moves the winner, what does that tell us?
- When should the system select none?

The production judge in one minute

A production selector might combine four kinds of evidence:

- Human calibration defines what usable means and supplies reference judgments.
- Visual diagnostics test object retention, structure, motion, and flicker.
- A VLM can test semantic condition match and describe failures.
- Hard safety rules veto failures that must never be averaged away.

A VLM is one signal. The judge is the rubric, evidence, thresholds, and decision rule together.

Thirty-to-forty-minute classroom route

Phase 1: Predict before scoring

Learners choose a candidate, give confidence from 1 to 5, record two visible reasons, and name one unacceptable failure.

Phase 2: Operationalize quality

Pairs define three observable criteria and at least one hard gate. Require operational language. 'Looks good' is not acceptable. 'Lane topology remains unchanged' is.

Phase 3: Apply the selector

Learners assign weights summing to 100, score eligible candidates, and compare the winner with the blind vote.

Phase 4: Stress-test the decision

Each pair creates a second mission profile, changes the weights, and reports the winner, runner-up, score margin, and ranking stability.

Phase 5: Transfer

Groups design a three-criterion selector and one hard gate for robotic manipulation, medical image generation, scientific surrogates, warehouse simulation, or code generation.

Core misconceptions

More samples guarantee improvement.	More samples enlarge the pool. Improvement also requires useful candidates and reliable selection.
The highest average is safe.	A critical defect can be hidden by high scores elsewhere. Gates come first.
A VLM is the judge.	A VLM provides evidence. The decision system also needs a rubric, thresholds, and rules.
Ray improves quality.	Ray changes throughput. It does not define quality.
Synthetic weather certifies AV safety.	It does not. This lesson teaches candidate triage and evaluation design.

Assessment and reuse

Exit ticket

1. What does N represent in Best-of-N?
2. Why should hard gates be applied before weighted scoring?
3. Name one way a misaligned selector could choose the wrong video.

Transfer challenge rubric

Mechanism	Cannot describe	Mentions samples	Separates generation and selection	Explains diversity, cost, and selection
Rubric	No criteria	Subjective only	Observable criteria	Operational criteria plus gates
Analysis	No selection	No calculation	Correct score	Sensitivity and explanation
Critique	Score as truth	Vague limit	Concrete failure	Calibration, audit, abstention

Suggested mastery: 9 of 12 points, with no zero.

Adaptation pattern

predict -> inspect -> formalize -> perturb -> diagnose -> transfer

- Introductory audience: hide the equation and production notes.
- Advanced audience: run notebooks/selector_lab.ipynb or source/scoring.py to study weight sensitivity and rating uncertainty.
- Different domain: replace the media, labels, criteria, and hard gates. Preserve the learning sequence.
- No network: serve the package locally. All core media and scripts are included.

Score provenance statement

Use this exact sentence when presenting:

The interaction uses fixed illustrative rubric scores. They were created to teach selector behavior and are not claims of autonomous-vehicle certification.